

NATIONAL ROCKETRY CHAMPIONSHIP

Manufacturing Review

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Introduction



- Our mission is to design, build, and fly a rocket that reaches 670 metres, delivers payloads successfully, and lands safely within the recovery zone.
- To achieve this, we aim to hit our target altitude with stable flight, deploy a reliable recovery system, and ensure the rocket is recovered within one kilometre of the launch site. We will also recover both payloads intact and successfully retrieve flight data from our avionics system.
- Throughout the flight, we will maintain stability and a controlled descent, supported by thorough ground testing before launch.
- Finally, we aim to recover the rocket in a reusable condition where possible, while fully complying with UKSEDS NRC safety regulations.



Project Plan

Task	Start Date	End Date	Completion	10/11/25	11/11/25	12/11/25	13/11/25	14/11/25	15/11/25	16/11/25	17/11/25	18/11/25	19/11/25	20/11/25	21/11/25	22/11/25	23/11/25	24/11/25	25/11/25	26/11/25	27/11/25	28/11/25	29/11/25	30/11/25	1/12/25
PDR	10/11/25	12/12/25	☒																						
Risk Assessment	30/11/25	11/12/25	☒																						
Deciding on Motor	29/11/25	03/12/25	☒																						
Team Logo	30/11/25	03/12/25	☒																						
Prototype Template	15/11/25	10/12/25	☒																						
Prototype Recovery Systems	29/11/25	05/12/25	☒																						
Structure and Material Selection	15/11/25	10/12/25	☒																						
Budget Allocation	10/11/25	07/12/25	☒																						
GANTT Chart	15/11/25	06/12/26	☒																						
December Exams	08/12/25	12/12/25	☒																						
Relevant Graphs	27/11/25	09/12/25	☒																						
Completing Slide Show	03/12/25	10/12/25	☒																						
Christmas Holiday	12/12/25	12/01/26	☒																						
CDR	13/12/25	19/02/26	☒																						
Team Structure and work allocation	11/10/2025	12/12/2025	☒																						
Finalise Design of Rocket	12/13/2026	4/20/2026	☒																						
Check Compliance of Design	12/13/2025	2/10/2026	☒																						
Proof of design functioning	2/19/2026	4/10/2026	☒																						
Finalise CAD Models	12/12/2025	2/10/2026	☒																						
Initial Avionics Design	13/12/25	10/02/26	☒																						
Exam	16/03/26	20/03/26	☒																						
Easter Holiday	20/03/26	01/05/26	☒																						
Manufacturing Review	19/02/26	01/05/26	☐																						
Flight Readiness	01/05/26	02/06/26	☐																						
Comp Day	17/06/26	24/06/26	☐																						
Exam	15/06/26	19/06/26	☐																						



Reason	Amount
2 x BMFA membership	£102.00
Motor	£30.00
Shock cords	£ 25.00
Parachute	£35.00
Mouser Electronics	£78.88
CPC Farnell Avionics	£25.49
Bambulab ABS-GF Filament	£31.99
Wires/Cabling	£50.96
Humidity Sensor + MicroSD	£19.30
Total	£398.82



Subsystem	Component	Estimated mass (g)	Margin	Notes
Propulsion	Cesaroni Pro29 108G68	120	-	Manufacturer value
	Motor mount + hardware	80	10%	3D printed + screws
Structures	Main airframe (balsa tube)	200	15%	Veneer construction
	Spruce fins (x4)	120	15%	Aero foil shaped
	Bulkheads / couplers	60	10%	plywood / printed
	Nosecone (PLA)	80	10%	3D printed
Recovery	24" parachute	70	10%	Spherachute
	Shock cords (kevlar)	40	10%	>2m
	Ejection charge hardware	20	10%	Charge wells, fittings
Avionics	Mercury V1 altimeter	25	-	Manufacturer value
	Rideshare PCB + IMU + SD	100	-	Estimated
	Battery (7.4V 2.25Ah)	140	-	Li-ion pack
	Wiring + switches	20	10%	Multi-core wiring
Payload	Customer Payload	75	Fixed	Competition defined
		955g		



Parameter	Value
Battery capacity	16.65 Wh
Customer Payload	5.63 Wh
Rideshare	~0.75 Wh
Total required	~6.4 Wh
Energy margin	>2.5×



Launch Vehicle Manufacture

Things to include:

- Motor configuration and retainment
- Recovery system configuration and assembly
- Interface with customer payload
 - How is it secured
 - How is it powered: directly from battery over buck converter (XT30 connector at the bottom) (possibly a video)
 - How is it assembled? (demo video preferred)



Recovery system configuration and assembly

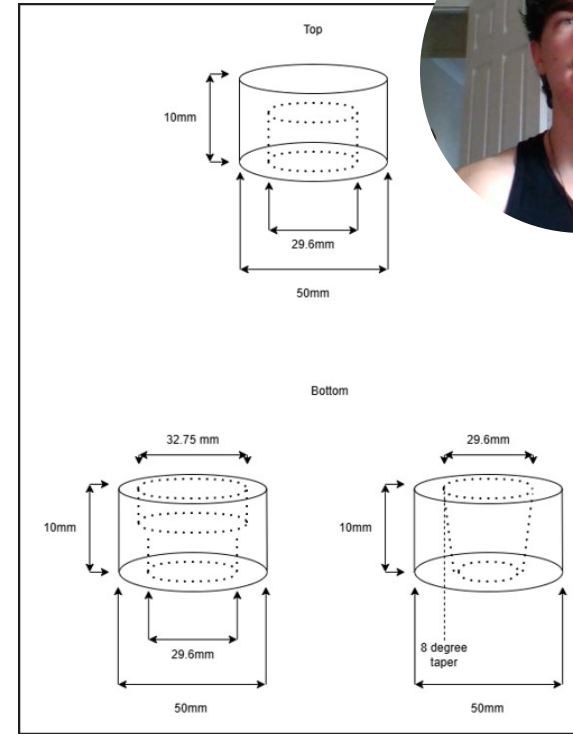


- We will be connecting our parachute to shocks cords which will be tied to eye bolts that will be screwed into the main body of the rocket.
- We are using flash paper as our source of pressure for the ejection charge and this will be controlled and lighted by an igniter. In order to control when the ejection charge is activated, we will use a flight computer to detect when our rocket reaches apogee.
- This follows the following guidelines:
 - 5.1 Use of Recovery System
 - 5.5 Flight Computer
 - 5.6 Isolation of Recovery Circuit
 - 5.8 Parachute Deployment
 - 5.9 Detection of Apogee
 - 5.10 Flight Computer Disarming Mechanism
 - 5.11 Transmission of Recovery Shock Loads
 - 5.12 Shock Cord



Motor Mount

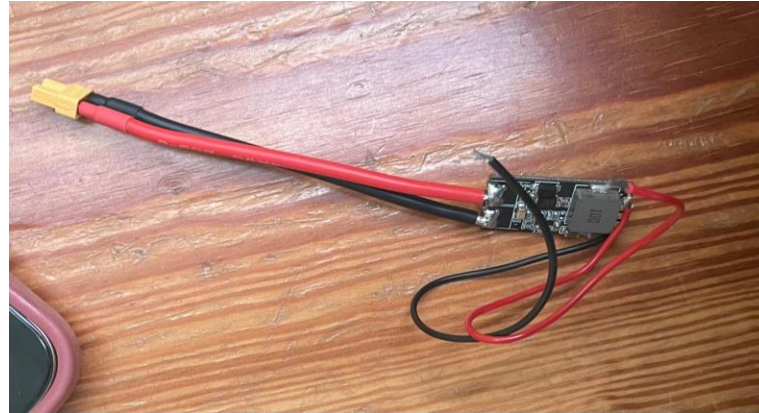
- Initial Motor Mount Scrapped
- New Design Created To Improve On Failures
- Rivets To Hold It In Place
- Testing:
 - Fixing to the rocket body
 - Vibration test
 - Compression stress test



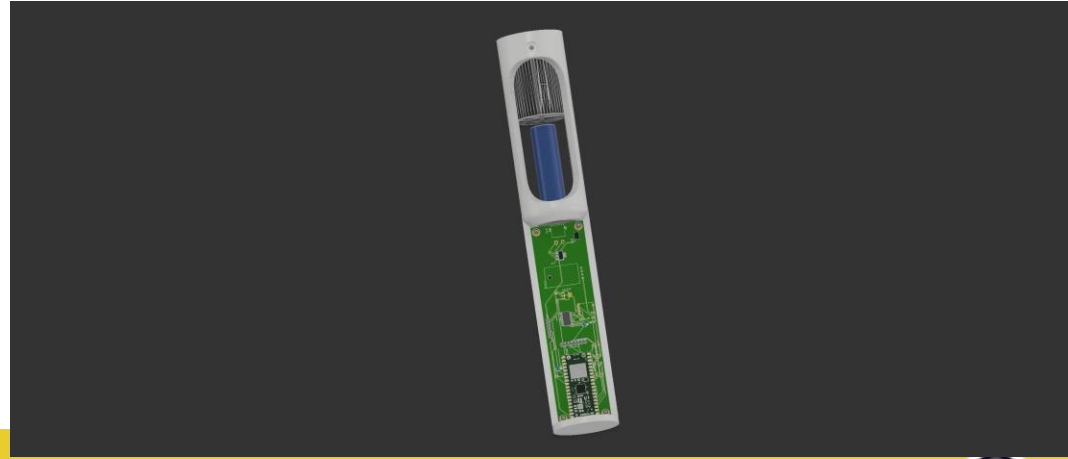
Interface with customer payload



- How is it secured: the avionics sled will be secured to the bottom of the payload bay with a bayonet mount, which is glued down.
- How is it powered: directly from battery over buck converter (XT30 connector at the bottom)

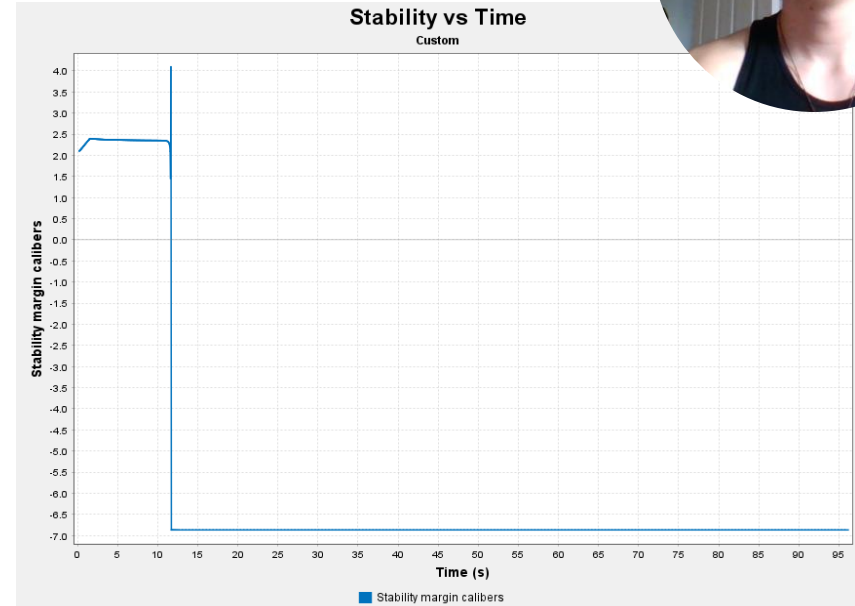


- How is it assembled?: it is secured to an avionics sled with the launch vehicle adapter directly connected to it



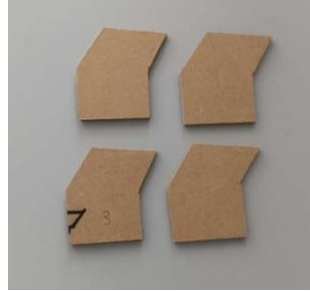
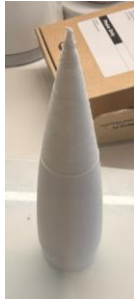
Launch Vehicle Simulation Data

- The windspeed was set to 0 m/s and the launch rod was set to 2 m when the simulations were ran.
- The simulations now show that the rocket reaches the desired apogee.
- The stability margin caliber is between 1.75 and 2.5 as required in 2.3.



Structures manufacturing

- Completed production:
 - Main body bought and tested
 - Fins material changed and tested
 - Nose cone designed and 3d printed



4 fins produced using saw machines and made of MDF wood panels

Design did not flex under an applied weight of 1.5 kg suspended both laterally and longitudinally



Main body made from cardboard tube

Design was able to support a 1.5kg weight without undergoing significant (>10mm) deflection



Rideshare Payload Manufacture

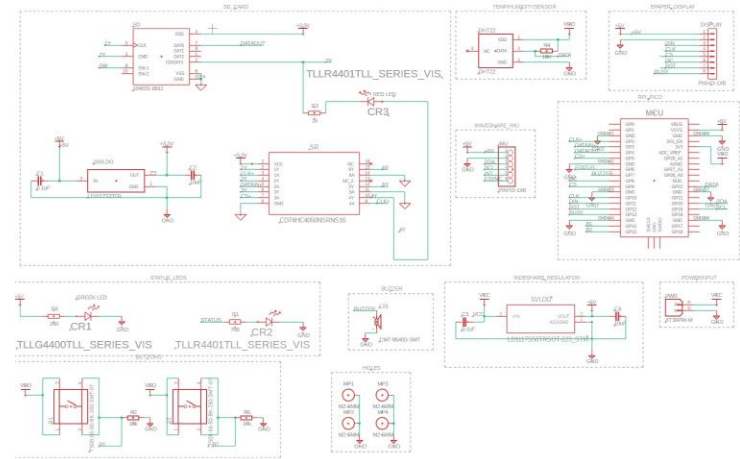
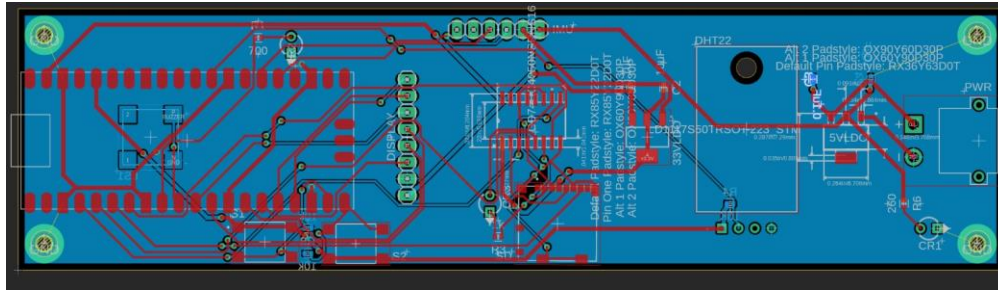


- General plan and progress on manufacturing:
 - In final stages of design
 - Need to get traces double checked
 - Then will send it off to JLCPCB
 - We will then solder the parts our self



Rideshare Payload Manufacture

- PCB manufacture:



Rideshare Payload Manufacture

- Assembly of rideshare and customer payload: both are going to be connected to the avionics sled
- We have tested individual parts of the PCB



System Engineering



Requirement ID	Met? [Yes, No, Partially]	Rational
NRC RS-TSN-MOR-001	Yes	The motor mount will transfer thrust being secured to both the body and the motor, allowing for minimal vibrations and prevention of slipping.
NRC RS-TSN-MOR-002	Partially	This is to be tested once we have all the parts
NRC RS-TSN-MOR-003	Partially	This is to be tested once we have all the parts
NRC RS-TSN-SMN-001	Yes	The rocket achieves a launch speed of 19.9m/s off the rail
NRC RS-TSN-SMN-003	Yes	The rocket has been designed to maintain a stability margin between 1.87 and 2.5 during all stages of flight
NRC RS-TSN-SAR-001	Partially	To be tested
NRC RS-TSN-SAR-002	Partially	To be tested
NRC RS-TSN-SAR-003	Partially	To be tested
NRC RS-TSN-SAR-004	Partially	To be tested
NRC RS-TSN-SAR-005	Yes	We are not using interconnected tubes in our design for the main body
NRC RS-TSN-SAR-006	Partially	To be tested once we have all the parts
NRC RS-TSN-SAR-007	Yes	Two rail guides are included in the main body of the launch vehicle
NRC RS-TSN-SAR-008	Yes	Finite Element Analysis was not used to select or test materials
NRC RS-TSN-ESS-001	Yes	The power of the battery to be used will be capable of supporting this.
NRC RS-TSN-ESS-002	Partially	This has not been tested; however, that is the theoretical expectation.
NRC RS-TSN-ESS-003	Yes	A multi-core wire has been included in the design
NRC RS-TSN-ESS-004	Yes	We're not communicating with the rocket wirelessly



System Engineering



NRCRS-TSN-RRS-001	Yes	Parachute to be used: Spherachute plus+ W/swivel 24". This has been simulated on OpenRocket.
NRCRS-TSN-RRS-002	Yes	OpenRocket simulation shows the system shall reduce the speed to 9.5m/s
NRCRS-TSN-RRS-003	Partially	Research is being done to estimate this
NRCRS-TSN-RRS-004	Yes	The Mercury V1 is UKRA approved.
NRCRS-TSN-RRS-005	Yes	A bulkhead is in between recovery system and electrical system
NRCRS-TSN-RRS-006	Yes	Flight computer is being used
NRCRS-TSN-RRS-007	Partially	Uncertainty of this for the moment. Awaiting part to confirm.
NRCRS-TSN-RRS-008	Yes	Holes to be created in the nozzle to secure the eye bolts using the nut.
NRCRS-TSN-RRS-007	Yes	We have a rde of shock cords to custimize the size
NRCRS-TSN-RRS-008	Yes	They're kevlar.
NRCRS-MRD-MIS-001	No	The OpenRocket simulation reached an apogee of just shy of 2200ft
NRCRS-MRD-MIS-002	No	We don't reach beyond the target height.
NRCRS-MRD-MIS-003	Yes	Theoretically impossible based on simulations
NRCRS-MRD-MIS-004	Yes	Motor fits in the guidelines
NRCRS-MRD-MIS-005	Yes	It's been purchased through UKSEDS
NRCRS-MRD-MIS-006	Yes	The motor form was submitted
NRCRS-MRD-MIS-007	Partially	To be tested
NRCRS-MRD-MIS-008	Partially	To be tested



UKSEDS
COMPETITIONS



System Engineering



NRCRS-MRD-CPD-001	Yes	5V is being used
NRCRS-MRD-CPD-002	Yes	The power of the battery supports the system for approximately 1 hour 30 minutes
NRCRS-MRD-CPD-003	Yes	Current of 1.5 amps will power the Customers Payload.
NRCRS-MRD-CPD-004	Yes	Position of port is bottom. Connectors and cables have arrived
NRCRS-MRD-CPD-005	Yes	Provided by the launch service
NRCRS-MRD-CPD-006	Yes	The adapter has been chosen and integrated into the Avionics Sled
NRCRS-MRD-CPD-007	Yes	Payload bay has measurements are 45mm diameter and 53mm length.
NRCRS-MRD-CPD-008	Yes	Design of rocket and integrated payload accounts for this
NRCRS-MRD-CPD-009	Partially	Design confirmed. Finals holes will be drilled when avionics sled is printed
NRCRS-MRD-CPD-010	Yes	Theres a hole in the rocket for the camera.
NRCRS-MRD-CPD-011	Yes	Bolts have arrived
NRCRS-MRD-RPD-001	Partially	Testing of single parts are complete. Awaiting PCB design verification
NRCRS-MRD-RPD-002	Yes	Using the barometer on a 10dof IMU
NRCRS-MRD-RPD-003	Partially	Individual parts are working. Screen works
NRCRS-MRD-RPD-004	Partially	Individual parts are working.
NRCRS-MRD-RPD-005	Partially	To be printed
NRCRS-MRD-RPD-006	Yes	Multiple sensor included in the rocket. Still deciding shall be used for this part of the requirement.



Testing

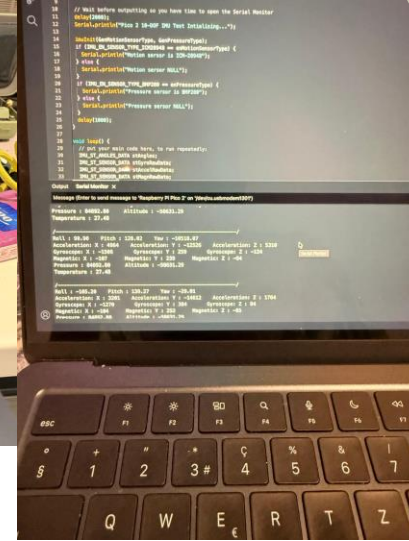
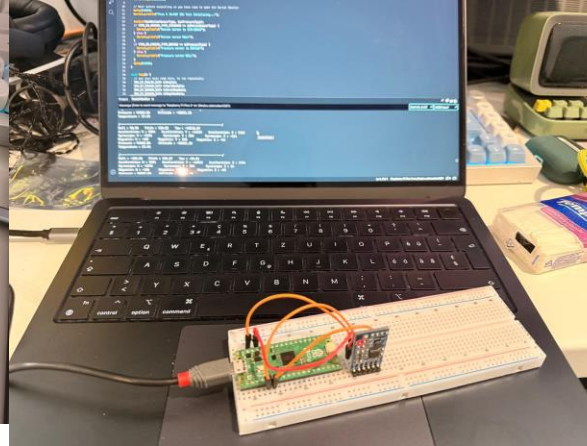
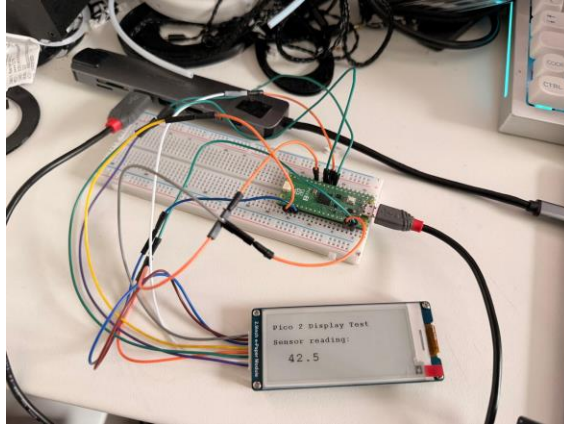
All Components have been tested individually and have met the required parameters needed for a safe and secure flight.



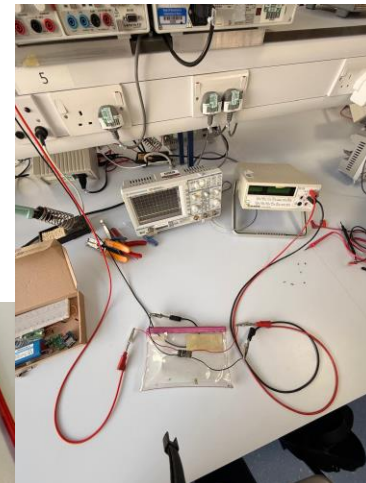
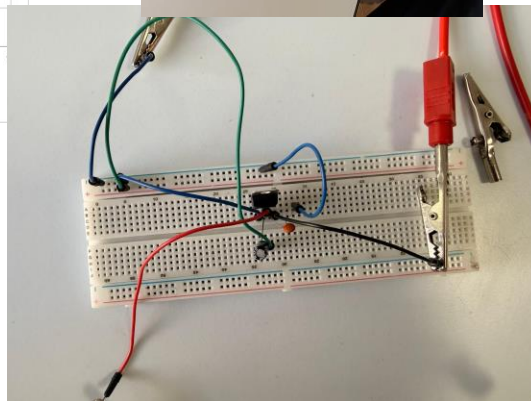
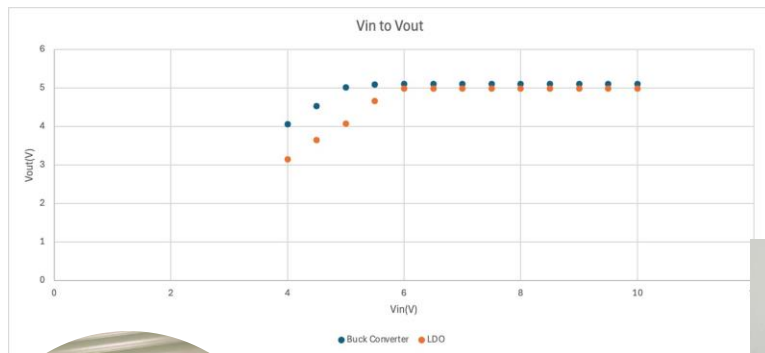
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TOTAL		955g		



Payload testing results



Payload testing results



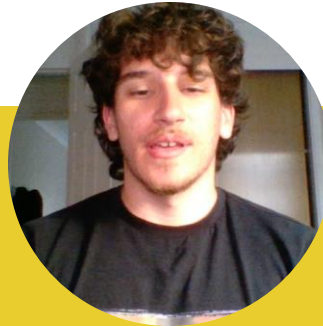
Risks & Next Steps

Risks :

- 1 - Motor Failure - Occasionally, a motor's casing can fail, leading to an explosion on the pad or in mid-air. This is why a minimum safety distance (usually 10-20 meters)
- 2 - Ballistic Re-entry - If the recovery system fails to deploy, the rocket will just freefall to the ground.
- 3 - Structural Failure - High-velocity flights exert massive G-forces. If the rocket fins aren't aligned or its airframe is weak, the rocket can "shred" or flutter, leading to an unpredictable flight path.
- 4 - Ignition Misfires - If the rocket doesn't launch after the countdown, the risk of a "hang-fire" (delayed ignition) is high
- 5 - Pyrotechnic Handling - during launch and manufacturing our team will be working with black powder or flash paper for ejection charges and recovery systems. Improper handling can cause premature ignition, which can lead to burns or eye injuries
- 6 - Launch Pad Stability - High winds (usually above 15-20 mph) can tip a launch, where the rocket veers into the wind and flies toward the spectators or away from the safe recovery zone.
- 7 - Hazardous Terrain - Rockets rarely land exactly where you want them. Recovery can lead you into high trees, water hazards, or near power lines

Next steps :

- 1 - manufacture motor casing
- 2 - review and test ejection system
- 3 - put rocket together



End Slide

